Jinx: Catching Duplicate Pull Requests from Coding Agents

[TODO: authors]

Table 1: Cross-author pairs (n = 342).

Strategy	R@1	R@5	MRR	False alarms % [95% CI]
Jinx (ours)	0.901	0.971	0.932	2.5 [1.9, 3.3]
All signals, LR	0.906	0.974	0.935	1.6 [1.1, 2.1]
Jinx + BGE rerank + Qwen-7B	0.901	0.974	0.931	2.8 [2.2, 3.5]
Claude Haiku 4.5 + Jinx	0.898	0.971	0.932	1.8 [1.3, 2.3]
Qwen2.5-14B + Jinx	0.871	0.971	0.917	2.9 [2.3, 3.7]
Claude Haiku 4.5 alone	0.842	0.971	0.902	5.4 [4.6, 6.1]
INTRUDE + time [2, 3]	0.810	0.953	0.873	4.6 [3.7, 5.7]
contrib-guard (bot)	0.778	0.918	0.843	11.7 [8.9, 14.6]
Links only	0.839	0.924	0.882	3.8 [3.0, 4.8]
TF-IDF [1]	0.757	0.921	0.832	9.4 [7.8, 10.9]
BGE-small embedding	0.763	0.915	0.830	6.4 [5.3, 7.5]

Abstract

Coding agents open many pull requests, and some repeat work that is already open. Jinx is a CPU-only GitHub Action and CLI that compares a new PR with the repository's open PRs and comments with ranked likely duplicates and plain reasons (shared files, overlapping changed lines, shared issue, similar text after removing agent boilerplate). On 1867 duplicate pairs from AIDev, Jinx ranks the true duplicate first for 0.901 of cross-author pairs, against 0.778 for the best existing duplicate-PR bot, with 2.5% false alarms, in about 22 ms per PR without an LLM. [TODO: video URL, repo URL, usage evidence]

ACM Reference Format:

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1 Introduction

[TODO: problem, existing tools (bots, commercial review tools), what Jinx adds]

2 Jinx

[TODO: architecture figure; template stripping; signals; logistic model; reasons; Action workflow; CLI example]

3 Evaluation

[TODO: head-to-head vs bots and classic detector (Table); latency; false alarms]

4 Usage and Limitations

[TODO: deployment evidence; limits: intent-level duplicates, silver labels]

5 Availability

[TODO: repo, PyPI `jinx-pr`, video, license MIT]

References

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